CHEVALLEY RESTRICTION THEOREM FOR CLASSICAL SIMPLE LIE ALGEBRA

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Abstract. In this article, I demonstrate the background knowledge , which is mainly about Lie group-Lie algebra correspondence and Chevalley restriction theorem and prove that it holds for the case of classical simple Lie groups.

1. Introduction

At the International Congress of Mathematicians in 1950, Claude Chevalley proposed the famous Chevalley restriction theorem, which states that the conjugate invariant functions on a semisimple Lie algebra are isomorphic to the Weyl group invariant functions on a Cartan subalgebra. The theorem was fully proved by Robert Steinberg in 1965 and generalized to semisimple algebraic groups. Since then, mathematicians have made many advances in the higher-dimensional [5] generalization of Chevalley's restriction theorem. When we study the properties of high-dimensional Hitchin morphism, the Chevalley restriction theorem plays a key role.

In this article, I demonstrate the background knowledge and prove that it holds true for the case of classical simple Lie groups.

1.1. Chevalley restriction theorem.

- G: a **complex connected semi-simple** Lie group;
- $\mathfrak{g}$: the Lie algebra of G.
- $\mathbb{C}[\mathfrak{g}]^G$: the polynomial functions on $\mathfrak{g}$ which are invariant under the **adjoint G-action**.
- $\mathfrak{h}$: a **Cartan subalgebra** of $\mathfrak{g}$.
- W: the **Weyl group** of G.
- $\mathbb{C}[\mathfrak{h}]^W$: the polynomial functions on $\mathfrak{h}$ which are invariant under the natural action of W.

Then restriction of polynomial functions induces an isomorphism as **graded algebra**:

$$\mathbb{C}[\mathfrak{g}]^G \cong \mathbb{C}[\mathfrak{h}]^W .$$

1.2. Elucidate.

- We introduce how to obtain the Lie algebra of a Lie group in the section of the Lie group–Lie algebra correspondence.
- Adjoint G-action (for linear Lie groups): We have a G-action on $\mathfrak{g}$ defined by $g.x = g^{-1}xg$. This induces an action on $\mathbb{C}[\mathfrak{g}]$, $g.P(x) := P(g^{-1}xg)$. In the section of the Lie group–Lie algebra correspondence, we will prove that this G-action is well - defined. P

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is a polynomial function, which means that when we have chosen a basis, the function P can be written as a polynomial of the coordinates. In summary,

$$\mathbb{C}[\mathfrak{g}]^G := \left\{ P : \mathfrak{g} \to \mathbb{C} \,\middle|\, \begin{array}{c} \text{for all } (x,g) \in \mathfrak{g} \times G, P\left(g^{-1}xg\right) = P(x) \\ P\left(\sum_{k=1}^n x_k e_k\right) = \sum a_{k_1..k_n} x_1^{k_1}..x_n^{k_n} \end{array} \right\}.$$

- For complex semisimple Lie algebra $\mathfrak{g}$, we have root space decomposition [1]

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha,$$

  while $\mathfrak{g}_\alpha := \{x \in \mathfrak{g} | \forall h \in \mathfrak{h}, \operatorname{ad} h(x) = \alpha(h)\, x\}$, $\alpha \in \mathfrak{h}^*$. And the root system Φ has a base Δ, which can span the whole space $\mathfrak{h}^*$.
- The killing form $\kappa(x,y) := \operatorname{tr}(\operatorname{ad}x \operatorname{ad}y)$ and its restriction on $\mathfrak{h}$ is non degenerate[1] , so for all $\alpha \in \mathfrak{h}^*$, there exists unique t_α such that

$$\alpha = \kappa(t_\alpha, -) : \mathfrak{h} \longrightarrow \mathbb{C}, \quad h \longmapsto \kappa(t_\alpha, h).$$

  Let $E := \mathbb{R}\alpha_1 \oplus \mathbb{R}\alpha_1 \oplus \cdots \oplus \mathbb{R}\alpha_l$, and $(\alpha, \beta) := \kappa(t_\alpha, t_\beta)$ then Φ becomes a root system of euclidean space[1] E.
- What is the Weyl group? Firstly, we define the Weyl group [1] $W := \langle \sigma_\alpha : \alpha \in \Phi \rangle \subseteq O(E)$, while $\sigma_\alpha : E \to E$ such that

$$\sigma_\alpha(\beta) = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha.$$

- What is its natural action? Let $\Delta = \{\alpha_1, ..., \alpha_l\} \subseteq \Phi$ be a base and fix a reflection $\sigma_\alpha : \alpha \in \Phi$, since $\sigma_\alpha(\Phi) = \Phi$, we have $\Delta \hookrightarrow \Phi$. We denote $\varphi : \alpha \mapsto t_\alpha$. Since Δ is a basis of $\mathfrak{h}^*$, there exists a unique $\tilde{\sigma}_\alpha$ that makes the following diagram commute.

$$\begin{array}{ccccc} \beta & & \Delta & \xhookrightarrow{\sigma_\alpha} & \Phi \\ \downarrow & & \varphi\downarrow & & \downarrow\varphi \\ t_\beta & & \mathfrak{h} & \overset{\tilde{\sigma}_\alpha}{\dashrightarrow} & \mathfrak{h} \end{array}$$

  Thus, we have W acts on $\mathfrak{h}$, extended by actions of its generators

$$W \times \mathfrak{h} \longrightarrow \mathfrak{h} \quad \left(\sigma_\alpha, \sum_i c_i t_{\alpha_i}\right) \longmapsto \sum_i c_i \tilde{\sigma}_\alpha(t_{\alpha_i}).$$

  In fact we can indentify $t_\alpha = \alpha$ via φ.

2. Lie group-Lie algebra correspondence

2.1. Lie algebra of linear Lie group.

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Lemma 2.1 (chap.3 in [2])**.** *Let $x, y \in M_n(\mathbb{C})$ then we have*
(i) (Trotter Product Formula)

$$\lim_{k\to\infty} \left(\exp\frac{x}{k}\exp\frac{y}{k}\right)^k = \exp(x+y).$$

(ii) (Commutator Formula)

$$\lim_{k\to\infty} \left(\exp\frac{x}{k}\exp\frac{y}{k}\exp\frac{-x}{k}\exp\frac{-y}{k}\right)^{k^2} = \exp(xy - yx).$$

Definition 2.2. Let $G \subseteq \mathrm{GL}_n(\mathbb{K})$ be a closed subgroup of $\mathrm{GL}_n(\mathbb{K})$, define

$$\mathbf{L}(G) := \{x \in M_n(\mathbb{K}) \,|\forall t \in \mathbb{R}, \exp(tx) \in G\}$$

with Lie bracket $[x, y] := xy - yx$, then $\mathbf{L}(G)$ be come a Lie algebra, called the Lie algebra of Lie group.

Proof. Using Lemma 2.1 we know $\mathbf{L}(G)$ is a Lie algebra with $[x, y] = xy - yx$. (Refer to [2]Chap.3 for a detailed proof) □

Proposition 2.3 (Chap.4 in [2])**.** *Let $\beta : V \times V \to \mathbb{K}$ be a bilinear map and V be a linear space of finite dimension over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, then we have a closed subgroup*

$$G = \mathrm{Aut}(V, \beta) := \{g \in \mathrm{GL}(V) : \forall v, w \in V, \beta(gv, gw) = \beta(v, w)\}$$

and its Lie algebra

$$\mathbf{L}G = \{x \in \mathfrak{gl}(V) : \beta(xv, w) + \beta(v, xw) = 0\}.$$

If $\beta(v, w) = v^\top Bw$ then

$$\mathbf{L}G = \left\{x \in \mathfrak{gl}(V) : x^\top B + Bx = 0\right\};$$

If $\beta(v, w) = v^* Bw$ then

$$\mathbf{L}G = \{x \in \mathfrak{gl}(V) : x^* B + Bx = 0\}.$$

Examples 2.4. Let

$$I_{p,q} := \begin{pmatrix} 0 & I_p \\ I_q & 0 \end{pmatrix}, \; J_n := \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

By Proposition 2.4

- If $\mathrm{O}_{p,q}(\mathbb{K}) := \left\{x \in \mathrm{GL}_{p+q}(\mathbb{K}) : x^\top I_{p,q} x = I_{p,q}\right\}$ then
$$\mathfrak{o}_{p,q}(\mathbb{K}) = \left\{x \in \mathfrak{gl}_{p+q}(\mathbb{K}) \,|x^\top I_{p,q} + I_{p,q} x = 0\right\} = \mathbf{L}\mathrm{O}_{p,q}(\mathbb{K});$$
- If$\mathrm{sp}_n(\mathbb{K}) := \left\{g \in \mathrm{GL}_{2n}(\mathbb{K}) : g^\top J_n + J_n g = 0\right\}$ then
$$\begin{aligned} \mathfrak{sp}_n(\mathbb{K}) :&= \left\{x \in \mathfrak{gl}_{2n}(\mathbb{K}) : x^\top J_n + J_n x = 0\right\} \\ &= \left\{\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathfrak{gl}_{2n}(\mathbb{K}) : B^\top = B, C^\top = C, D = -A^\top\right\}; \end{aligned}$$
- Let $G := \mathrm{SO}_{p,q}(\mathbb{C}) := \mathrm{O}_{p,q}(\mathbb{C}) \cap \mathrm{SL}_n(\mathbb{C})$, then we have
$$\mathfrak{so}_{p,q}(\mathbb{C}) := \mathbf{L}\mathrm{SO}_{p,q}(\mathbb{C}) = \left\{x \in \mathfrak{gl}_{p+q}(\mathbb{C}) \,|x^\top I_{p,q} + I_{p,q} x = 0\right\} = \mathfrak{o}_{p,q}.$$

2.2. **The Lie functor.**

Proposition 2.5. *Let $\varphi : G_1 \to G_2$ be a morphism of Lie groups, then there exists unique morphism of Lie algebras $\mathbf{L}(\varphi)$ making the following diagram commute*

$$\begin{tikzcd} G_1 \arrow[r, "\varphi"] & G_2 \\ \mathbf{L}(G_1) \arrow[u, "\exp_{G_1}"] \arrow[r, dashed, "\mathbf{L}(\varphi)"] & \mathbf{L}(G_2) \arrow[u, "\exp_{G_2}"'] \end{tikzcd}$$

And we have a functor $\mathbf{L}$ *from the category of linear Lie groups to the category of Lie algebras over* $\mathbb{K}$.

Proof. (Refer to [2] Chap.3 for a detailed proof) For the existence of $\mathbf{L}(\varphi)$, we define

$$\mathbf{L}(\varphi) : \mathbf{L}(G_1) \longrightarrow \mathbf{L}(G_2) \qquad x \longmapsto \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \varphi\left(\exp_{G_1}(tx)\right).$$

□

Examples 2.6 (**Adjoint G-action**)**.** For linear Lie groups, let $g \in G$, then we have an isomorphism of Lie groups:

$$c_g : G \longrightarrow G,\ h \longmapsto ghg^{-1}.$$

Then it is easy to know

$$L(c_g) : \mathbf{L}(G) \longrightarrow \mathbf{L}(G),\ x \longmapsto gxg^{-1}.$$

which is an isomorphism of Lie algebras. This induces a smooth action of the Lie group on $\mathbf{L}(G)$

$$G \times \mathbf{L}G \longrightarrow \mathbf{L}G \quad (g, x) \longmapsto gxg^{-1}.$$

2.3. **Adjoint Representation.** .

Lemma 2.7. *Let* $x, y \in \mathfrak{gl}(V)$ *be two arbitrary linear maps, then we have*

$$e^x y e^{-x} = e^{\operatorname{ad} x} y.$$

Proof. Let $\lambda_x : y \longmapsto xy$, $\rho_x = y \longmapsto yx$, which are commutative, then we have

$$e^x y e^{-x} = e^{-\rho_x} e^{\lambda_x} y = e^{\lambda_x - \rho_x} y.$$

□

Proposition 2.8. *We define the adjoint representation of* G *by*

$$\mathrm{Ad} : G \longrightarrow \mathrm{Aut}(G), \quad g \longmapsto c_g.$$

Then we have

$$\mathbf{L}(\mathrm{Ad}) = \mathrm{ad} : \mathbf{L}G \longrightarrow \mathbf{L}\mathrm{Aut}(G) \subseteq \mathfrak{gl}(\mathbf{L}G).$$

Proof. Since $e^x y e^{-x} = e^{\operatorname{ad} x} y$, we have the following commutative diagram

$$\begin{tikzcd}
G \arrow[r, "\mathrm{Ad}"] & \mathrm{Aut}(G) \\
\mathbf{L}G \arrow[u, "\exp_1"] \arrow[r, "\mathrm{ad}"'] & \mathfrak{gl}(\mathbf{L}G) \arrow[u, "\exp_2"']
\end{tikzcd}$$

For uniqueness we know $\mathbf{L}(\mathrm{Ad}) = \mathrm{ad}$. □

Definition 2.9 (Inner isomorhism of $\mathfrak{g}$)**.** For the linear Lie group over $\mathbb{C}$, define

$$\mathrm{Inn}(\mathfrak{g}) := \left\langle e^{\operatorname{ad} x} : x \in \mathfrak{g} \right\rangle = \left\langle \mathrm{Ad}\left(e^{-x}\right) : x \in \mathfrak{g} \right\rangle.$$

Remark 2.10. The group $\mathrm{Inn}(\mathfrak{g})$ acting on $\mathfrak{g}$ can identify as some adjoint G-actions because of

$$\mathrm{Inn}(\mathfrak{g}) \subseteq \{\mathrm{Ad}(g) : g \in G\}.$$

Corollary. *If H is a normal Lie subgroup of G, then $\mathbf{L}H$ is an ideal of $\mathbf{L}G$.*

Proof. $H \lhd G$ implies that $\mathrm{Ad} : G \to \mathrm{Aut}(H)$, so we have $\mathrm{ad} : \mathbf{L}G \to \mathfrak{gl}(\mathbf{L}H)$. So for all $x \in \mathbf{L}G, \mathrm{ad}\, x$ preserves $\mathbf{L}H$. □

3. Classification of complex Simple Lie algebra and their Weyl groups

The materials in the section are well-known and readers should refer to [1] for more details.

Lemma 3.1. *We have a canonical isomorphism as Lie groups*

$$\mathrm{SO}_{n,n}(\mathbb{C}) \cong \mathrm{SO}_{2n}(\mathbb{C}).$$

In fact $\mathrm{SO}_{n,n}(\mathbb{C}) \to \mathrm{SO}_{2n}(\mathbb{C}),\ g \mapsto ugu^{-1}$ *gives an isomorphism with*

$$u = \begin{pmatrix} I_n & I_n \\ -iI_n & iI_n \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & iI \end{pmatrix} \begin{pmatrix} I & I \\ -I & I \end{pmatrix}, \quad i^2 = -1.$$

Proof: Calculate and verify directly.

Remark 3.2. Using the Lie functor, we have $\mathfrak{so}_{n,n}(\mathbb{C}) \cong \mathfrak{so}_{2n}(\mathbb{C})$ as Lie algebras.

Lemma 3.3 (The classification theorem [1] for semisimple Lie algebras)**.** *Every finite-dimensional semisimple Lie algebra over $\mathbb{C}$ is isomorphic to a direct sum of simple Lie algebras. These simple Lie algebras can be classified into the following types: A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, and G_2. The general forms of these Lie algebras are given by:*

- *A_n $(n \geq 1)$: the Lie algebra $\mathfrak{sl}_{n+1}(\mathbb{C})$, with complex Lie group $\mathrm{SL}_{n+1}(\mathbb{C})$;*
- *B_n $(n \geq 2)$: the Lie algebra*
$$\mathfrak{so}_{2n}(\mathbb{C}) \simeq \mathfrak{so}(\mathbb{C}^{2n+1}, T) := \{x \in \mathfrak{gl}_{2n+1}(\mathbb{C}) : x^\top T + Tx = 0\};$$
while $T := \begin{pmatrix} I_{n,n} & 0 \\ 0 & 1 \end{pmatrix} \in M_{2n+1}(\mathbb{C})$
- *C_n $(n \geq 3)$: the Lie algebra*
$$\mathfrak{g} = \mathfrak{sp}_{2n}(\mathbb{C}) = \{x \in \mathfrak{gl}_{2n}(\mathbb{C}) : x^\top J_n + J_n x = 0\};$$
- *D_n $(n \geq 4)$: the Lie algebra*
$$\mathfrak{so}_{2n}(\mathbb{C}) \simeq \mathfrak{so}(\mathbb{C}^{2n}, I_{n,n}) := \{x \in \mathfrak{gl}_{2n}(\mathbb{C}) : x^\top I_{n,n} + I_{n,n} x = 0\};$$
- *E_6: a complex Lie algebra of dimension 78;*
- *E_7: a complex Lie algebra of dimension 133;*
- *E_8: a complex Lie algebra of dimension 248;*
- *F_4: a complex Lie algebra of dimension 52;*
- *G_2: a complex Lie algebra of dimension 14.*

Definition 3.4 (Cartan subalgebra)**.** We say that the subalgebra $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$ if $\mathfrak{h}$ is nilpotent and $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$.

Lemma 3.5 (Chap.6 in [2])**.** *Let $\mathfrak{g}$ be a simisimple Lie algebra over $\mathbb{C}$, then the following are equivalent:*

- *$\mathfrak{h}$ is a Cartan subalgebra.*
- *$\mathfrak{h}$ is a maximal torus subalgebra.*
- *$\mathfrak{h}$ is a maximal abelian subalgebra consisting of semisimple elements.*

Lemma 3.6 ([2]Cartan subalgebra of classical simple Lie algebra)**.** *Let*

(1) $\mathfrak{h}$ *denote one of Cartan subalgebra of* $\mathfrak{g}$*;*
(2) $\varepsilon_j : \mathfrak{h} \to \mathbb{C}$ *be the linear functions defined by*

$$\varepsilon_k(\mathrm{diag}(h, -h)) := h_k, \quad h = (h_1, ..., h_n);$$

(3) Δ *be the basis of the root system;*
(4) $E_{i,j}$ *denote the elementry matrix and* $E_k := E_{kk}$ *.*

Then we have we have the following results.

- A_n *(*$n \geq 1$*):When* $\mathfrak{g} = \mathfrak{sl}_{n+1}(\mathbb{C})$

$$\mathfrak{h} = \mathrm{span}\{E_j - E_{j+1} : j = 1, \ldots, n\} = \{\mathrm{diag}(h, -\mathrm{tr}\, h) : h \in \mathbb{C}^n\}.$$

  The corresponding root system and one of its bases are

$$A_{n-1} := \{\varepsilon_j - \varepsilon_k : 1 \leq j, k \neq 1 \neq n-1\};$$

$$\Delta = \{\varepsilon_j - \varepsilon_{j+1} | j = 1, ..., n\}.$$

- B_n *(*$n \geq 2$*): When* $\mathfrak{g} = \mathfrak{so}_{2n+1}(\mathbb{C})$

$$\mathfrak{h} = \mathrm{span}\{E_j - E_{n+j} : j = 1, \ldots, n\} = \{\mathrm{diag}(h, -h, 0) : h \in \mathbb{C}^n\}.$$

  The corresponding root system and one of its bases are

$$B_n := \{\pm\varepsilon_j, \pm\varepsilon_j \pm \varepsilon_k : j, k = 1, \ldots, n, j \neq k\};$$

$$\Delta = \{\varepsilon_j - \varepsilon_{j+1} | j = 1, ..., n-1\} \cup \{\varepsilon_n\}.$$

- C_n *(*$n \geq 3$*): When* $\mathfrak{g} = \mathfrak{sp}_{2n}(\mathbb{C})$

$$\mathfrak{h} = \mathrm{span}\{E_j - E_{n+j} : j = 1, \ldots, n\} = \{\mathrm{diag}(h, -h) : h \in \mathbb{C}^n\}.$$

  The corresponding root system and one of its bases are

$$C_n := \{\pm 2\varepsilon_j, \pm\varepsilon_j \pm \varepsilon_k : j, k = 1, \ldots, n, j \neq k\};$$

$$\Delta = \{\varepsilon_j - \varepsilon_{j+1} | j = 1, ..., n-1\} \cup \{2\varepsilon_n\}.$$

- D_n *(*$n \geq 4$*): When* $\mathfrak{g} = \mathfrak{so}_{2n}(\mathbb{C})$

$$\mathfrak{h} = \mathrm{span}\{E_j - E_{n+j} : j = 1, \ldots, n\} = \{\mathrm{diag}(h, -h) : h \in \mathbb{C}^n\}.$$

  The corresponding root system and one of its bases are

$$D_n := \{\pm\varepsilon_j \pm \varepsilon_k : j, k = 1, \ldots, n, j \neq k\};$$

$$\Delta = \{\varepsilon_j - \varepsilon_{j+1} | j = 1, ..., n-1\} \cup \{\varepsilon_{n-1} + \varepsilon_n\}.$$

Proof. The following proof is based on the example of type A_n case and C_n case, as the proofs for other types are similar.

The A_{n-1} case : It is easy to know $\mathfrak{h} = \{\mathrm{diag}(a_1, ..., a_n) | a_1 + ... + a_n = 0\}$ and

$$(\mathrm{ad}h)(E_{ij}) = (h_i - h_j) E_{ij}, \quad h = \mathrm{diag}(h_1, ..., h_n).$$

Further more

$$\mathfrak{sl}_n = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathbb{C} E_{ij}.$$

Then we can work out the corresponding root system is

$$A_{n-1} := \{\varepsilon_j - \varepsilon_k : 1 \leq j, k \neq 1 \neq n-1\};$$

For C_n case, using

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathfrak{sp}_n(\mathbb{C}) \quad \Longleftrightarrow \quad d = -a^\top,\, b^\top = b,\, c^\top = c,$$

we see that $\mathfrak{g} := \mathfrak{sp}_n(\mathbb{C})$ has an abelian subalgebra

$$\mathfrak{h} = \operatorname{span}\{E_j - E_{n+j} : j = 1, \dots, n\} = \{\operatorname{diag}(h, -h) : h \in \mathbb{C}^n\}.$$

If $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in Z_{\mathfrak{g}}(\mathfrak{h})$, then for all $h = \operatorname{diag}(h_1, ..., h_n) \in M_n(\mathbb{C})$

$$0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} h & \\ & -h \end{pmatrix} - \begin{pmatrix} h & \\ & -h \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ah - ha & -bh - hb \\ ch + hc & hd - dh \end{pmatrix}.$$

$ah = ha$ implies that a is a diagonal matrix. $hb + bh = 0$ implies that $b = 0$. So $Z_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. Similarly, some of $E_{j,n+k} + E_{k,n+j}$ and $E_{n+k,j} + E_{n+j,k}$ form a basis of all $\mathfrak{g}_\alpha$.

Then we can work out the corresponding root system is

$$C_n := \{\pm 2\varepsilon_j, \pm\varepsilon_j \pm \varepsilon_k : j, k = 1, \dots, n, j \neq k\}.$$

□

Lemma 3.7 (Generators of Weyl group)**.** *Let Δ be a base of Φ, then (refer to* [1]*) the Weyl group*

$$W := \langle \sigma_\alpha : \alpha \in \Phi \rangle = \langle \sigma_\alpha : \alpha \in \Delta \rangle$$

Lemma 3.8. *If the following conditions hold:*

(1) E_1 is a subspace of E_2 as a Euclidean space such that $E_2 = E_1 \oplus E_1^\perp$, when $E_1^\perp = \mathbb{R}\varepsilon$.
(2) H_1 subgroup of $O(E_1)$ and H_2 subgroup of $O(E_2)$.
(3) For all $\sigma \in H_2, \sigma(\varepsilon) = \varepsilon$.
(4) $H_1 = \{\sigma|_{E_1} : \sigma \in H_2\}$.

Then $H_1 \simeq H_2$.

Proof. There is a surjective homomorphism $\varphi : H_2 \longrightarrow H_1,\ \sigma \longmapsto \sigma|_{E_1}$. Its kernel is

$$\operatorname{Ker}(\varphi) = \{\sigma \in H_2 : \sigma|_{E_1} = \mathrm{id}_{E_1}, \sigma(\varepsilon) = \varepsilon\} = \{\mathrm{id}\}.$$

Thus, it is an isomorphism. □

Lemma 3.9 ([4]Weyl group of classical simple Lie algebra)**.** *Let W be the Weyl group of $\mathfrak{g}$, then we have the following results.*

- *Type A_{n-1} case: $W \simeq S_n$ with natural action ($E_j := E_{jj}$) generated by*

$$\sigma_{\varepsilon_k - \varepsilon_{k+1}}(E_j) = E_{\pi(j)}, \pi = (k, k+1) \in S_n.$$

- *Type B_n case: $W \simeq S_n \ltimes \mathbb{Z}^n$ with natural action ($e_j := E_{jj} - E_{n+j,n+j}$) generated by*

$$\sigma_{\varepsilon_k - \varepsilon_{k+1}}(e_j) = e_{\pi(j)}, \pi = (k, k+1) \in S_n;$$

$$\sigma_{\varepsilon_n}(e_n) = -e_n.$$

- *Type C_n case: $W \simeq S_n \ltimes \mathbb{Z}^n$ with natural action ($e_j := E_{jj} - E_{n+j,n+j}$) generated by*

$$\sigma_{\varepsilon_k - \varepsilon_{k+1}}(e_j) = e_{\pi(j)}, \pi = (k, k+1) \in S_n;$$

$$\sigma_{2\varepsilon_n}(e_n) = -e_n.$$

- *Type D_n case: $W \simeq S_n \ltimes \mathbb{Z}^{n-1}$ with natural action ($e_j := E_{jj} - E_{n+j,n+j}$) generated by*

$$\sigma_{\varepsilon_k - \varepsilon_{k+1}}(e_j) = e_{\pi(j)}, \pi = (k, k+1) \in S_n;$$
$$\sigma_{\varepsilon_n + \varepsilon_{n-1}}(e_j) = -e_j, \quad j \in \{n-1, n\}.$$

Proof. Type A_{n-1} case: We claim that

$$\kappa(x, y) = 2n\mathrm{tr}(xy), \forall x, y \in \mathfrak{h}$$

Let $h = \mathrm{diag}(h_1, ..., h_n) \in M_n(\mathbb{C})$, for $(\mathrm{ad}h)^2(E_{ij}) = (h_i - h_j)^2 E_{ij}$ we know

$$\begin{aligned}
\kappa(h, h) &= \sum_{i \neq j} (h_i - h_j)^2 = 2\sum_{i<j} (h_i - h_j)^2 \\
&= 2\sum_{i<j} (h_i - h_j) h_i + 2\sum_{i<j} (h_j - h_i) h_j \\
&= 2\sum_{i=1}^{n-1} h_i \sum_{j=i+1}^{n} (h_i - h_j) + 2\sum_{j=2}^{n} h_j \sum_{i=1}^{j-1} (h_j - h_i) \\
&= 2\sum_{i=1}^{n} h_i \sum_{j=i}^{n} (h_i - h_j) + 2\sum_{i=1}^{n} h_i \sum_{j=1}^{i-1} 2(h_i - h_j) \\
&= \sum_{i=1}^{n} h_i \sum_{j=1}^{n} 2(h_i - h_j) = 2n\sum_{i=1}^{n} h_i^2 = 2n\mathrm{tr}\left(h^2\right).
\end{aligned}$$

By $\kappa(x, y) = \frac{1}{4}[\kappa(x+y, x+y) - \kappa(x-y, x-y)]$, our claim holds. More precisely

$$\sigma_\alpha(E_j) = e_j - \frac{2\mathrm{tr}(E_j t_\alpha)}{\mathrm{tr}({t_\alpha}^2)} t_\alpha, \quad t_{\varepsilon_i} = E_i - E_{n+1}.$$

Then we can work out the action, and conclude that

$$H_2 := \left\langle \sigma_{e_k - e_{k+1}} | k = 1, ..., n-1 \right\rangle \simeq \langle (k, k+1) \in S_n | k = 1, ..., n-1 \rangle = S_n,$$

and let $\varepsilon = e_1 + e_2 + ... + e_n, E_1 = \varepsilon^\perp = \mathfrak{h}$, by 3.8 we have

$$W = \{\sigma|_{E_1} : \sigma \in H_2\} \simeq H_2 \simeq S_n.$$

The type D_n, B_n case is similar to C_n case. Let us consider the C_n case and identify $\mathfrak{g} = \mathfrak{sp}_n(\mathbb{C})$ as a subalgebra of $\mathfrak{sl}_{2n}(\mathbb{C})$, the killing form

$$\kappa(x, y) = 4n\mathrm{tr}(xy), \forall x, y \in \mathfrak{h}$$

then we can work out all reflections in a similar way. □

4. Chevalley restriction theorem

In the section of Introduction, we have stated the Chevalley restriction theorem.

Theorem 4.1 (Chevalley restriction theorem [3]). *The restriction of polynomial functions induces an isomorphism as **graded algebra**:*

$$\mathbb{C}[\mathfrak{g}]^G \cong \mathbb{C}[\mathfrak{h}]^W \qquad P \longmapsto P|_{\mathfrak{h}}.$$

Proposition 4.2 (Type A_{n-1} case)**.** *When $G = \mathrm{SL}_n$, $\mathfrak{g} = \mathfrak{sl}_n$, the induced map is an isomorphism.*

Proof. We have known that $\mathfrak{h} = \{\mathrm{diag}(a_1, ..., a_n) | a_i \in \mathbb{C}, a_1 + ... + a_n = 0\}$. Let $E_{i,j}$ denote the elementry matrix and $E_k := E_{kk}$. Then polynomial function means

$$P\left(\sum_{i\neq j} x_{i,j}E_{ij} + \sum_{k=1}^{n} x_k E_{kk}\right)$$

are polynomials of $x_{i,j}$ and x_k with the equation

$$x_1 + x_2 + ... + x_n = 0.$$

Then we can identify the restriction map as

$$P\left(\sum_{i\neq j} x_{i,j}E_{ij} + \sum_{k=1}^{n} x_k E_{kk}\right) \longmapsto P|_{\mathfrak{h}} = P\left(\sum_{i\neq j} 0E_{ij} + \sum_{k=1}^{n} x_k E_{kk}\right).$$

Obviously the induced map is a homomorphism of graded algebra, the proof of this proposition is mainly divided into the following steps

Step 1: $\mathbb{C}[\mathfrak{h}]^W = \mathbb{C}[\sigma_2, ..., \sigma_n]$, while $\sigma_1, \sigma_2, ..., \sigma_n$ are elementary symmetric polynomials with equation $\sigma_1 = 0$.

Proof. The action $W = S_n$ acting on $\mathbb{C}[\mathfrak{h}]$, is defined by

$$\pi.P\left(\sum_{k=1}^{n} x_k E_{kk}\right) = P\left(\sum_{k=1}^{n} x_{\pi(k)} E_{kk}\right), \quad \pi \in S_n.$$

According to basic theorem of elementary symmetric polynomial, we have

$$\mathbb{C}[\mathfrak{h}]^W \subseteq \mathbb{C}[\sigma_1, \sigma_2, ..., \sigma_n].$$

For $\sigma_1 = 0$, $\mathbb{C}[\mathfrak{h}]^W = \mathbb{C}[\sigma_2, ..., \sigma_n]$. □

Step 2: $\mathbb{C}[\tau_2, ..., \tau_n] \subseteq \mathbb{C}[\mathfrak{g}]^G$, while $\tau_2, .., \tau_n$ are some polynomials such that

$$\det\left(tI_n - \sum_{i\neq j} x_{i,j}E_{ij} - \sum_{k=1}^{n} x_k E_{kk}\right) = t^n - 0t^{n-1} + \tau_2 t^{n-2} - \tau_3 t^{n-3} + ... + (-1)^n \tau_n.$$

Proof. It is welled known that

$$\det\left(tI - g^{-1}Ag\right) = \det\left(tg^{-1}(I - A)g\right) = \det\left(tI - A\right).$$

Thus, the polynomial coefficients are invariant under the adjoint G-action, which implies that they are invariant polynomials. □

Step 3: The induced map to $\mathbb{C}[\mathfrak{h}]^W$ is well defined.

Proof. We can identify the restriction map as

$$P\left(\sum_{i\neq j} x_{i,j}E_{ij} + \sum_{k=1}^{n} x_k E_{kk}\right) \longmapsto P|_{\mathfrak{h}} = P\left(\sum_{i\neq j} 0E_{ij} + \sum_{k=1}^{n} x_k E_{kk}\right).$$

So if $P \in \mathbb{C}[\mathfrak{g}]^G$, then

$$P\left(g^{-1}Ag\right) = P(A) \Rightarrow P\left(\sum_{i\neq j} x_{i,j} g^{-1}E_{ij}g + \sum_{k=1}^{n} x_k g^{-1}E_{kk}g\right) = P\left(\sum_{i\neq j} x_{i,j}E_{ij} + \sum_{k=1}^{n} x_k E_{kk}\right).$$

Let $x_{i,j} = 0$, the equation still holds, so we have $P|_{\mathfrak{h}} \in \mathbb{C}[\mathfrak{h}]^G$. According to the well-known linear algebra conclusion, these two matrices are conjugate

$$\sum_{k=1}^{n} x_{\pi(k)}E_{kk}\ , \sum_{k=1}^{n} x_k E_{kk}.$$

So we have $P|_{\mathfrak{h}} \in \mathbb{C}[\mathfrak{h}]^G \subseteq \mathbb{C}[\mathfrak{h}]^W$. □

Step 4(Surjective):

Proof. In the equation

$$\det\left(tI_n - \sum_{i\neq j} x_{i,j}E_{ij} - \sum_{k=1}^{n} x_k E_{kk}\right) = t^n - 0t^{n-1} + \tau_2 t^{n-2} - \tau_3 t^{n-3} + ... + (-1)^n \tau_n.$$

Let $x_{i,j} = 0$, we have $\tau_i|_{\mathfrak{h}} = \sigma_i \forall i = 2, ..., n$, which generate $\mathbb{C}[\mathfrak{h}]^W$. Obviously

$$P(\tau_2, ..., \tau_n)|_{\mathfrak{h}} = P(\tau_2|_{\mathfrak{h}}, ..., \tau_n|_{\mathfrak{h}}) = P(\sigma_2, ..., \sigma_n).$$

Thus $\operatorname{Im}\varphi = \mathbb{C}[\mathfrak{h}]^W$. □

Step 5(Injective): It suffices to show that $\operatorname{Ker}\varphi = 0$.

Proof. Let $P \in \ker\varphi$, then $P(\mathfrak{h}) = \{0\}$ and for all $(x, g) \in \mathfrak{g} \times G$

$$P(g^{-1}xg) = P(x).$$

We denote

$$\mathfrak{g}^s = \{g^{-1}hg | g \in \mathrm{GL}_n,\ h \in \mathfrak{h}\} = \{g^{-1}hg | g \in \mathrm{SL}_n,\ h \in \mathfrak{h}\}$$

the set of diagonalizable elements. Then we have $P(\mathfrak{g}^s) = \{0\}$. For P is a continuous function, it suffices to show that $\mathfrak{g}^s$ is dense in $\mathfrak{g}$.

Let $x \in \mathfrak{g}$, we construct a sequence $\{x_k\}$ in $\mathfrak{g}^s$ such that $\lim_{k\to\infty} x_k = x$. We know there exists $g \in G$ such that

$$g^{-1}xg = \begin{pmatrix} J(\lambda_1, n_1) & & & \\ & J(\lambda_2, n_2) & & \\ & & \ddots & \\ & & & J(\lambda_r, n_r) \end{pmatrix}, \quad J(\lambda_j, n_j) = \begin{pmatrix} \lambda_j & 1 & & & \\ & \lambda_j & 1 & & \\ & & \ddots & & \\ & & & \lambda_j & 1 \\ & & & & \lambda_j \end{pmatrix}_{n_j\times n_j}.$$

So let

$$g^{-1}x_k g = \begin{pmatrix} A_{k,1} & & & \\ & A_{k,2} & & \\ & & \ddots & \\ & & & A_{k,r} \end{pmatrix}, \quad A_{k,j} = \begin{pmatrix} \lambda_j & 1 & & & \\ & \lambda_j & 1 & & \\ & & \ddots & & \\ & & & \lambda_j & 1 \\ \frac{1}{k} & & & & \lambda_j \end{pmatrix}_{n_j\times n_j},$$

then for all j, we have

$$\lim_{k\to\infty} A_{k,j} = J(\lambda_j, n_j).$$

Thus

$$\lim_{k\to\infty} x_k = \lim_{k\to\infty} g \begin{pmatrix} A_{k,1} & & \\ & \ddots & \\ & & A_{k,r} \end{pmatrix} g^{-1} = g \begin{pmatrix} J(\lambda_1, n_1) & & \\ & \ddots & \\ & & J(\lambda_r, n_r) \end{pmatrix} g^{-1} = x,$$

and the characteristic polynomials

$$\det(\lambda I - A_{k,j}) = \begin{vmatrix} \lambda-\lambda_j & -1 & & & \\ & \lambda-\lambda_j & -1 & & \\ & & \lambda-\lambda_j & & \\ & & & \ddots & -1 \\ \frac{1}{k} & & & & \lambda-\lambda_j \end{vmatrix}_{n_j\times n_j} = (\lambda-\lambda_j)^{n_j} - \frac{1}{k}$$

do not have multiple roots, so $A_{k,j}$ is diagonalizable, so does x_k. This proof ends. □

In summary, we have proved that the induced map is an isomorphism. □

Corollary (Generators)**.** *When $G = \mathrm{SL}_n$, $\mathfrak{g} = \mathfrak{sl}_n$, we have found out all generators of invariant polynomials. In other words $\mathbb{C}[\mathfrak{g}]^G = \mathbb{C}[\tau_2, .., \tau_n]$.*

Proof. If there a polynomial function $P : \mathbb{C}^{n-1} \to \mathbb{C}$ such that $P(\sigma_2, ..., \sigma_n) = 0$, then for all $(z_2, ..., z_n) \in \mathbb{C}^{n-1}$, consider the polynomial equation

$$T^n - 0T^{n-1} + z_2T^{n-2} - z_3T^{n-2} + ... + (-1)^n z_n = 0,$$

the equation have n solution $t_1, ..., t_n$, by Vieta theorem so the equations

$$\sigma_1(x_1, ..., x_n) = 0, \sigma_2(x_1, ..., x_n) = z_2, ..., \sigma_n(x_1, ..., x_n) = z_n$$

have a solution $(x_1, ..., x_n) = (t_1, ..., t_n)$, this means

$$P(z_2, ..., z_n) = 0, \forall (z_2, ..., z_n) \in \mathbb{C}^{n-1}.$$

Thus $P = 0$. We have proved that the generators $\sigma_2, ..., \sigma_n$ are algebraically independent. Similarly, we can prove $\tau_2, .., \tau_n$ are algebraically independent using the characteristic polynomial. □

Proposition 4.3 (Type B_n Case)**.** *When $G = \mathrm{SO}_{2n+1}(\mathbb{C})$, the induced map is surjective. In addition,*

$$\mathbb{C}[\mathfrak{h}]^W = \mathbb{C}[\sigma_1, ..., \sigma_n], \quad \sigma_k(x_1, ..., x_n) := \sum_{i_1<...<i_k} (x_{i_1}...x_{i_k})^2$$

Proof. We have known that in this case $\mathfrak{g} = \mathfrak{so}_{2n+1}(\mathbb{C})$ and one of Cartan subalgebra is

$$\mathfrak{h} = \mathrm{span}\{E_j - E_{n+j} : j = 1, \ldots, n\} = \{\mathrm{diag}(h, -h, 0) : h \in \mathbb{C}^n\}.$$

And the Weyl group $W \simeq S_n \ltimes \mathbb{Z}^n$ with natural action ($e_j := E_{jj} - E_{n+j,n+j}$) generated by

$$\sigma_{\varepsilon_k - \varepsilon_{k+1}}(e_j) = e_{\pi(j)}, \pi = (k, k+1) \in S_n;$$

$$\sigma_{\varepsilon_n}(e_n) = -e_n;$$

Let us identify $x \in \mathfrak{h}$ as $x = (x_1, ..., x_n) \in \mathbb{C}$, then

$$\sigma_{\varepsilon_n} P(x_1, ..., x_n) = P(x_1, ..., -x_n);$$

$$\sigma_{\varepsilon_k-\varepsilon_{k+1}}P\left(x_1,...,x_n\right)=P\left(x_{\pi(1)},...,x_{\pi(n)}\right),\pi=(k,k+1)\in S_n.$$

So

$$\mathbb{C}\left[\mathfrak{h}\right]^W\subseteq\mathbb{C}\left[x_1^2,...,x_n^2\right]^{S_n}=\mathbb{C}\left[\sigma_1,...,\sigma_n\right],\quad \sigma_k\left(x_1,...,x_n\right):=\sum_{i_1<...<i_k}\left(x_{i_1}...x_{i_k}\right)^2.$$

In addition, the generators σ_i are coefficients of (with $h=$diag$(x_1,...,x_n)$)

$$\det\left(tI-\begin{pmatrix}h & & \\ & -h & \\ & & 0\end{pmatrix}\right)=t\left(t^2-x_1^2\right)...\left(t^2-x_n^2\right).$$

Similarly

$$\det\left(tI-\begin{pmatrix}a & b & y\\ c & -a^T & z\\ y & z & f\end{pmatrix}\right)=t\sum_{k=0}^{n}\left(-1\right)^k\tau_k t^{2n-2k}$$

is G-invariant and we have τ_k is G-invariant such that $\tau_k|\mathfrak{h}=\sigma_k$. Thus, the induced map is surjective. □

Proposition 4.4 (Type C_n case)**.** *When $G=\mathrm{sp}_{2n}(\mathbb{C})$, the induced map is surjective. In addition,*

$$\mathbb{C}\left[\mathfrak{h}\right]^W=\mathbb{C}\left[\sigma_1,...,\sigma_n\right],\quad \sigma_k\left(x_1,...,x_n\right):=\sum_{i_1<...<i_k}\left(x_{i_1}...x_{i_k}\right)^2.$$

Proof. It is entirely similar to the proof of type B_n case. □

Proposition 4.5 (Type D_n case)**.** *When $G=\mathrm{SO}_{2n}(\mathbb{C})$, the induced map is surjective. In addition,*

$$\mathbb{C}\left[\mathfrak{h}\right]^W=\mathbb{C}\left[\sigma_1,...,\sigma_{n-1},x_1x_2..x_n\right],\quad \sigma_k\left(x_1,...,x_n\right):=\sum_{i_1<...<i_k}\left(x_{i_1}...x_{i_k}\right)^2.$$

Proof. Let us identify $x\in\mathfrak{h}$ as $x=(x_1,...,x_n)\in\mathbb{C}$, then

$$\sigma_{\varepsilon_n+\varepsilon_{n-1}}P\left(x_1,...,x_{n-1},x_n\right)=P\left(x_1,...,-x_{n-1},-x_n\right);$$

$$\sigma_{\varepsilon_k-\varepsilon_{k+1}}P\left(x_1,...,x_n\right)=P\left(x_{\pi(1)},...,x_{\pi(n)}\right),\pi=(k,k+1)\in S_n.$$

So

$$\mathbb{C}\left[x_1^2,...,x_n^2\right]^{S_n}=\mathbb{C}\left[\sigma_1,...,\sigma_n\right]\subseteq\mathbb{C}\left[\mathfrak{h}\right]^W,\quad \sigma_k\left(x_1,...,x_n\right):=\sum_{i_1<...<i_k}\left(x_{i_1}...x_{i_k}\right)^2.$$

While $x_1x_2...x_n$ are also a invariant polynomial, we can vertify that $\mathbb{C}\left[\mathfrak{h}\right]^W=\mathbb{C}\left[\sigma_1,...,\sigma_{n-1},x_1x_2..x_n\right]$. In addition, the generators σ_i $(i\leq n-1)$ are coefficients of (with $h=$diag$(x_1,...,x_n)$)

$$\det\left(tI-\begin{pmatrix}h & \\ & -h\end{pmatrix}\right)=\left(t^2-x_1^2\right)...\left(t^2-x_n^2\right).$$

Similarly, using the coefficient of characteristic polynomial, we have $\tau_i(1\leq n-1)$ are G-invariant polynomials such that $\tau_i|_{\mathfrak{h}}=\sigma_i(1\leq n-1)$. According to 3.1 , we know $\exists u,\forall x\in\mathfrak{g},(uxu^{-1})^\top=-uxu^{-1}$, and the determinant of skew-sysmmetric matrix is the square of Pfaffian. So $\det x=\mathrm{Pf}\left(uxu^{-1}\right)^2$. Thus

$$\mathrm{Pf}\left(uxu^{-1}\right)=\sqrt{\det x}\in\mathbb{C}\left[\mathfrak{g}\right]^G,$$

and its restriction on $\mathfrak{g}$ is $(\sqrt{-1})^n x_1x_2...x_n$. □

Definition 4.6 (rank and the regular element)**.** Let us denote the space

$$\mathfrak{g}_x^a := \mathfrak{g}^a(\operatorname{ad} x) := \bigcup_{n\geqslant 1} \operatorname{Ker}(a\mathrm{id} - \operatorname{ad} x)^n.$$

(1) The number
$$\operatorname{rank}(\mathfrak{g}) := \min\left\{\dim \mathfrak{g}^0(\operatorname{ad} x) \mid x \in \mathfrak{g}\right\}.$$
is called the **rank** of $\mathfrak{g}$.
(2) An element $x \in \mathfrak{g}$ is called **regular** if $\dim \mathfrak{g}^0(\operatorname{ad} x) = \operatorname{rank}(\mathfrak{g})$.
(3) We write $\mathfrak{g}^r$ for the set of regular elements in $\mathfrak{g}$ and $\mathfrak{g}^s$ for the set of semisimple elements in $\mathfrak{g}$.

Remark 4.7. Since $\dim \mathfrak{g}^0(\operatorname{ad} x)$ is the multiplicity of 0 as a root of the characteristic polynomial

$$\det(\operatorname{ad} x - t\mathrm{id}_{\mathfrak{g}}) = \sum_{k=0}^{n} p_k(x)t^k.$$

of $\operatorname{ad} x$, we have

$$\operatorname{rank}(\mathfrak{g}) = \min\{k \in \mathbb{N} \mid p_k \neq 0\}.$$

Let $\{e_j, 1 \leq j \leq n\}$ be a basis of $\mathfrak{g}$ and $[e_i, e_j] = \sum_{k=1}^n a_{i,j}^k e_k, \quad x = \sum_{i=1}^n x_i e_i$ then

$$\operatorname{ad} x(e_j) = [x, e_j] = \sum_{i=1}^n x_i[e_i, e_j] = \sum_{i=1}^n \sum_{k=1}^n a_{i,j}^k x_i e_k = \sum_{k=1}^n \left(\sum_{i=1}^n a_{i,j}^k x_i\right) e_k.$$

So the entries of the matrix of $\operatorname{ad} x$ are linear combination of x_i, which implies that all the functions $p_k : \mathfrak{g} \to \mathbb{C}$ are polynomials.

Since the characteristic polynomial is invariant under adjoint action, we can immediately conclude the following.

Lemma 4.8 (Chap.6 in [2])**.** *The polynomial $p_r(x) \in \mathbb{C}[\mathfrak{g}]^G$ and*

$$\mathfrak{g}^r = \{x \in \mathfrak{g} | p_r(x) \neq 0\}.$$

In addtion, $\mathfrak{g}^r$ is an open, G-stable, dense and connected subset of $\mathfrak{g}$.

Proof. The proof of this lemma about dense and connected in the reference book [2] uses the results of Exercise 6.1.2 and 6.1.2. Next, I will prove the results of these two exercises.

To show that the set is dense in $\mathfrak{g}$, equivalently, $Z := \{x \in \mathbb{C}^n : p(x) = 0\}$ has no interior points, where p is a non-constant polynomial. Otherwise, if $p(x) = 0, \forall x \in B(x_0, r)$, then using the Taylor expansion at $x = x_0$,

$$p(x) = \sum_{|\alpha|=k}^{\deg p} \frac{\partial^\alpha p(x_0)}{\alpha!}(x - x_0)^\alpha, \quad \alpha = (\alpha_1, .., \alpha_n),$$

we can obtain $p = 0$.

To show that the set $X = \mathfrak{g}^r = \{x \in \mathfrak{g} \mid p(x) \neq 0\}$ is connected, we consider the properties of complex algebraic varieties and their complements.

First, recall that $\mathbb{C}^n$ is homeomorphic to $\mathbb{R}^{2n}$, which is connected. The zero set of a non-constant polynomial $p(x)$, denoted $Z = \{x \in \mathbb{C}^n \mid p_r(x) = 0\}$, is a proper algebraic subset of $\mathbb{C}^n$. Specifically, Z is a complex hypersurface of complex dimension $n - 1$, which corresponds to real dimension $2n - 2$ in $\mathbb{R}^{2n}$.

In topology, removing a subset of codimension greater than 1 from a manifold does not disconnect the manifold. Since Z has real codimension 2 in $\mathbb{R}^{2n}$, removing Z from $\mathbb{R}^{2n}$ does not disconnect it. Therefore, the complement $X = \mathbb{C}^n \setminus Z$ remains connected.

Therefore, the set $X = \mathfrak{g}^r$ is connected. □

Lemma 4.9 (Chap.6 in [2])**.** *Let $\mathfrak{g}$ be semisimple Lie algebra over $\mathbb{C}$ and $\mathfrak{h}$ be a Cartan subalgebra , then*

(1) If $\Delta = \Delta(\mathfrak{g}, \mathfrak{h})$ be a base, then

$$\mathfrak{g}^r \cap \mathfrak{h} = \mathfrak{h} \setminus \bigcup_{\alpha \in \Delta} \mathrm{Ker}\alpha \neq \emptyset.$$

(2) For all $x \in \mathfrak{g}^r$, the space $\mathfrak{g}_x^0$ form a Cartan subalgebra.
(3) For all $x \in \mathfrak{g}^r \cap \mathfrak{h}$, we have $\mathfrak{h} = \mathfrak{g}_x^0$.
(4) Every regular element in $\mathfrak{g}$ is simisimple. In other words

$$\mathfrak{g}^r \cap \mathfrak{g}^s = \mathfrak{g}^r.$$

Corollary. *Let H be the set of Cartan subalgebras of $\mathfrak{g}$, then*

$$\mathfrak{g}^r \subseteq \bigcup_{\mathfrak{h} \in H} \mathfrak{h}.$$

Proof. If $x \in \mathfrak{g}^r$, then $x \in \mathfrak{g}_x^0 = \mathfrak{h}$, which is a Cartan subalgebra. □

Lemma 4.10 (Chap.6 in [2])**.** $\mathrm{Inn}(\mathfrak{g})$ *acts transitively on the set of Cartan subalgebra of $\mathfrak{g}$. In other words, if we fixed a Cartan subalgebra, then*

$$H = \{g\mathfrak{h} | g \in \mathrm{Inn}(\mathfrak{g})\} \Rightarrow \mathfrak{g}^r \subseteq \bigcup_{g \in \mathrm{Inn}(\mathfrak{g})} g\mathfrak{h} \subseteq \bigcup_{g \in G} g\mathfrak{h}g^{-1}.$$

Proposition 4.11 (Injective)**.** *For general complex connected Lie group, the induced map is an injection.*

Proof. Let $P \in \mathbb{C}[\mathfrak{h}]^W$ such that $P|_{\mathfrak{h}} = 0$, then we have

$$\forall x \in \bigcup_{g \in G} g\mathfrak{h}g^{-1},\ P(x) = 0.$$

By the lemma above, we have

$$\forall x \in \mathfrak{g}^r, P(x) = 0, \quad \mathfrak{g}^r = \{p_r(x) \neq 0 | x \in \mathfrak{g}\}.$$

But 4.8 we have known that $\mathfrak{g}^r$ dense in $\mathfrak{g}$, which implies $P = 0$.

In summary , we have proved the kernel of the induced map is 0. □

5. Conclusion and Future Work

I have proved that the Chevalley restriction theorem holds true for the case of classical simple Lie groups, and worked out the concrete generators of the invariant polynomial functions in $\mathbb{C}[\mathfrak{g}]^G$ and $\mathbb{C}[\mathfrak{h}]^W$.

In the future, I will strive to calculate the generators in the Chevalley restriction theorem for exceptional complex simple Lie algebras, and prove that the induced map is also a surjection for general complex semisimple Lie groups.

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